FORMULAS 101

Formula Syntax

All formulas start with an ***equals sign***

Arguments are always surrounded by ***parentheses***

Arguments are separated by ***commas*** *in the US, but other regions may use different list separators (like* ***semi-colons****)*

= **MATCH**(lookup_value, lookup_array, [match_type])

The **function name** tells Excel what type of operation you're about to perform *(Excel offers ~500 functions)*

Note: Function names aren't case-sensitive, and aren't always required; basic arithmetic and logical operations often don't need one:

- *= A1 + B1*
- *= A1 / B1*
- *= A1 > B1*
- *= A1 = B1*

These are **arguments**, which vary by function and provide Excel with the info needed to evaluate a result

Note: Not all arguments are required; optional arguments are surrounded by square brackets *(like* ***[match_type]*** *above)*

Most functions have at least one required argument, but some don't require any, like **ROW()**, **COLUMN()**, **TODAY()** or **NOW()**

PRO TIP:

=MATCH(A2,

MATCH(lookup_value, **lookup_array**, [match_type])

As you begin writing a formula, the ***Function ScreenTips*** *box will guide you through each individual argument – this is an extremely helpful tool!*

MAVEN ANALYTICS

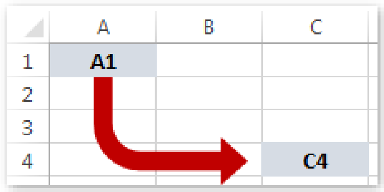
Reference Types

Reference types allow you to "recycle" formulas across multiple cells, without having to manually update your references (which would be completely impractical)

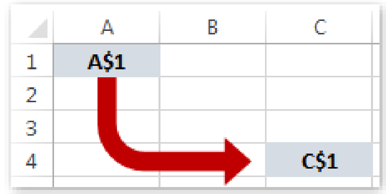
Cell references are **relative** by default (A1). This **allows the reference to change** as the formula is copied to new cells

The **$** symbol is used to create **fixed** references. You can fix entire cells (A1) or just the column ($A1) or row (A$1), which **prevents references from changing** as the formula is copied to new cells

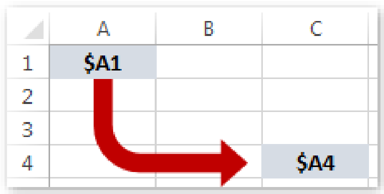
Relative Column & Row



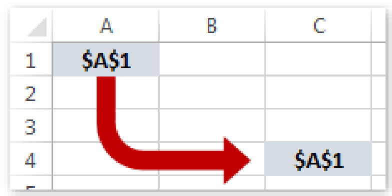
Relative Column, Fixed Row



Fixed Column, Relative Row



Fixed Column & Row



PRO TIP:

*Mastering reference types is my **#1 tip** for working efficiently with formulas*

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Common Errors

Error Type	What it means	How to fix it
######	Column isn't wide enough to display values	*Drag or double-click column border to increase width, or right-click to set custom column width*
#NAME?	Excel does not recognize text in a formula	*Make sure that function names are correct, references are valid and spelled properly, and quotation marks and colons are in place*
#VALUE!	Formula has the wrong type of argument	*Check that your formula isn't trying to perform an arithmetic operation on text strings or cells formatted as text*
#DIV/0!	Formula is dividing by zero or an empty cell	*Check the value of your divisor; if 0 is correct, use an IF statement to display an alternate value if you choose*
#REF!	Formula refers to a cell that is not valid	*Make sure that you didn't move, delete, or replace cells that are referenced in your formula*
#N/A	Formula can't find a referenced value	*Check that all references and formula arguments evaluate properly (the most common cause is a lookup value with no match)*

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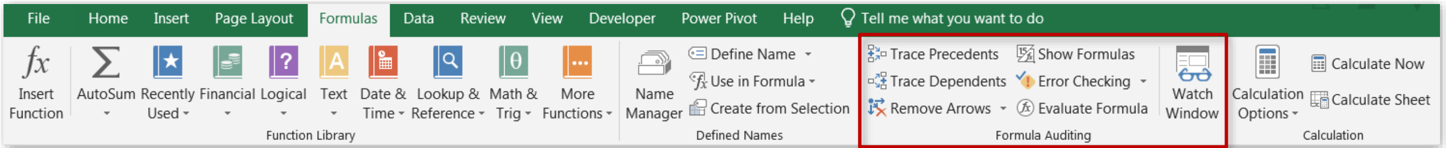
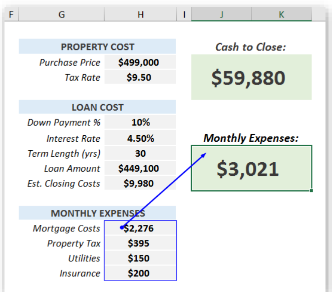


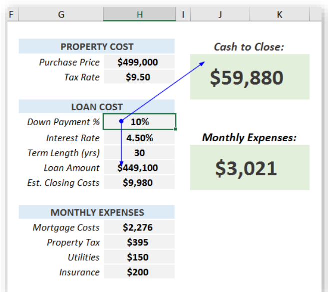
Trace Precedents

Identifies cells which **affect** the value of the one selected



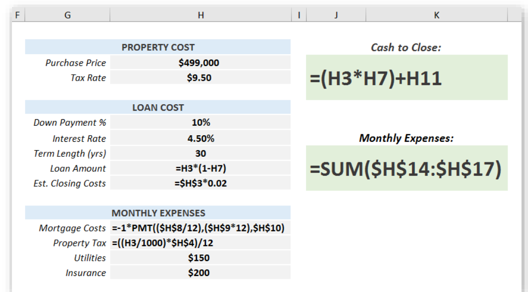
Trace Dependents

Identifies cells which **are affected by** the value of the one selected



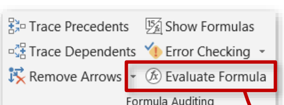
Show Formulas

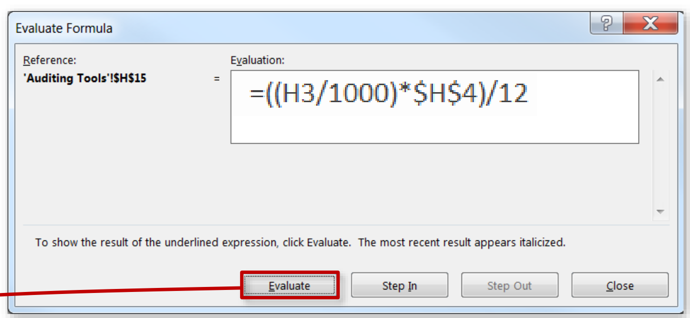
Temporarily displays all formulas in the worksheet as **text strings**



TIP: *To select all cells containing formulas, use **Ctrl-G** to launch the Go-To menu, then select **Special > Formulas***

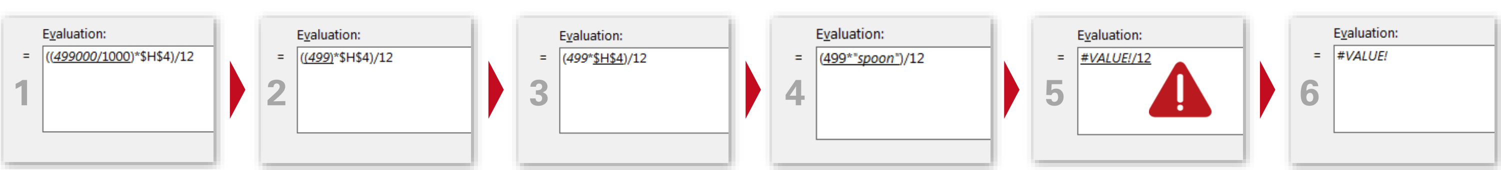
Auditing Tools





Evaluate Formula

Allows you to cycle through each individual calculation step within a formula, see how each component evaluates, and pinpoint the source of the error



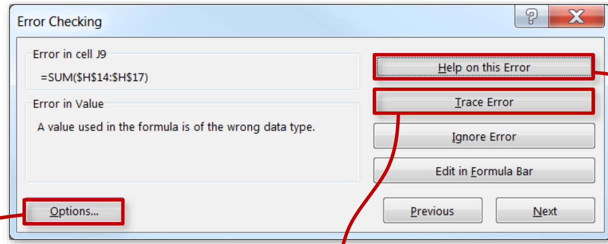
PRO TIP:

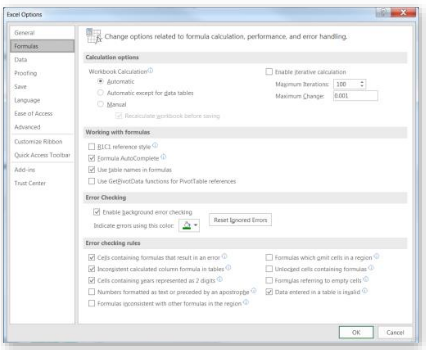
*Evaluate Formula is my **go-to tool** for breaking down complex or unfamiliar formulas*

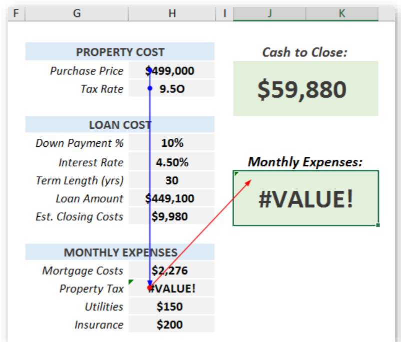
Auditing Tools

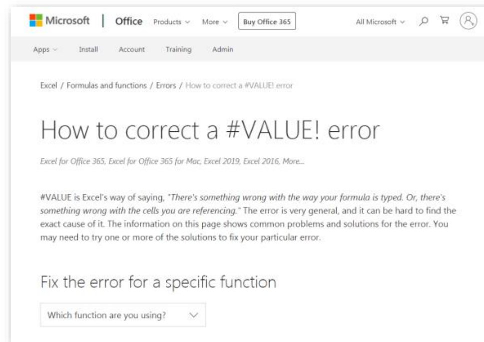
Error Checking

Scans the sheet for errors and provides a summary with options to trace the source, ignore the error, modify your options, or link out to Microsoft support









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CTRL Shortcuts

Ctrl + Arrow

- **Jumps to the last cell** in a data region, in the direction of the arrow

Ctrl + Shift + Arrow

- **Selects to the last cell** in a data region, in the direction of the arrow

Ctrl + Home/End

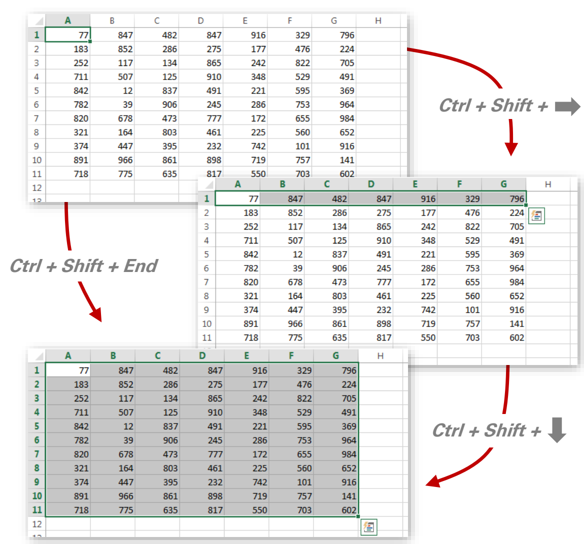
- Jumps to the **Home** (*top-left*) or **End** (*bottom-right*) cell in a region

Ctrl + .

- Jumps straight to **each corner** within a selected cell range

Ctrl + PgUp/PgDn

- **Switches worksheet tabs**, either to the **left** (*PgUp*) or **right** (*PgDn*)



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FULL LIST: https://exceljet.net/keyboard-shortcuts

Fn Shortcuts

F1

- Launches the **Excel help** pane *(default)*
- Links to the **Microsoft Support** website *(tool-specific)*

F2

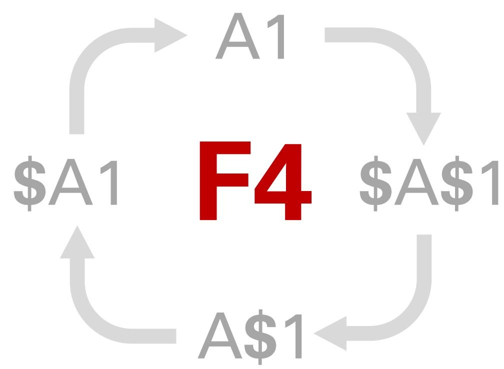
- Allows you to **edit** the active cell
- Highlights cells referenced by the active formula

F4

- Repeats the **last action** taken *(default)*
- Toggles **absolute/relative** cell references within a formula

F9

- Calculates all workbook formulas *(when in **manual** mode)*
- Evaluates **each function argument** within the formula bar



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FULL LIST: https://exceljet.net/keyboard-shortcuts

Common Mac Shortcuts

Mac Shortcut	Purpose	PC Equivalent
Command-T	*Cycles between cell reference types*	**F4**
Command-Y	*Repeats the last user action*	**F4**
Control-U	*Displays cell ranges tied to a given formula*	**F2**
Command-Arrow	*Jumps to the edge of a contiguous data array*	**CTRL-ARROW**
Command-Shift-Arrow	*Extends a selection to the edge of a data array*	**CTRL-SHIFT-ARROW**
Command-Fn-Up/Down	*Jumps between workbook tabs*	**CTRL-PAGE UP/DOWN**

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FULL LIST: https://exceljet.net/keyboard-shortcuts

Alt Key Tips

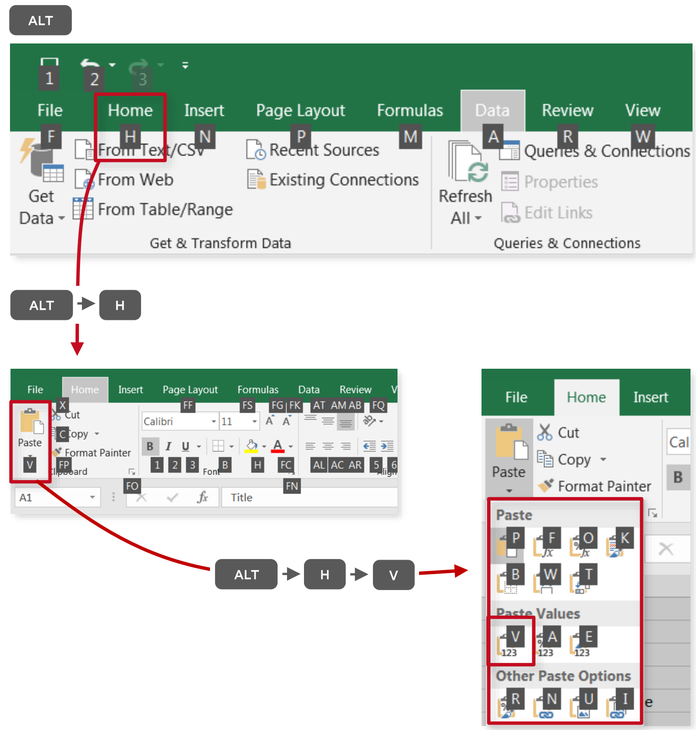
- Allow you to quickly access tools from ribbon menus and sub-menus using only the **keyboard** *(no clicks!)*
- Each keystroke takes you a layer deeper, until you land on the tool you need *(**Note**: Simply **press & release** the Alt key, instead of holding it down)*
- There are hundreds of combinations, so start by focusing on the tools that you use most frequently:

Some of my go-to key tips

Alt – H – V – V
- Paste Special as Values

Alt – A – T
- Add or remove filters

Alt – M – V
- Evaluate Formula



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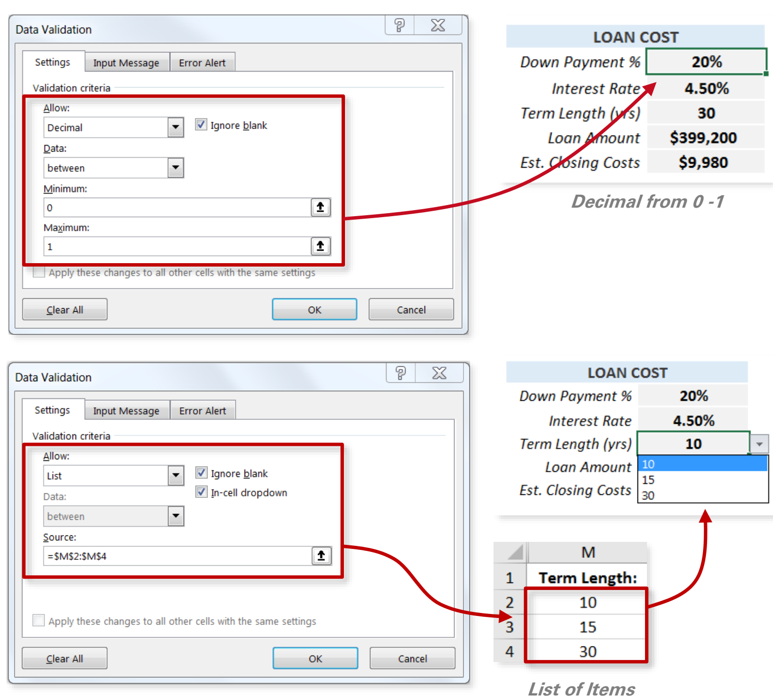
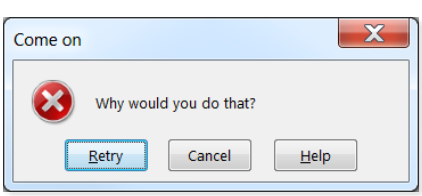
Data Validation

Restricts the values that a user can enter into a given cell, based on:

- **Number Type** *(Whole vs. Decimal)*
- **Value** *(Between, Less Than, Equal To, etc)*
- **List of Items** *(Based on cell range or manual list)*
- **Date/Time** *(Between, Less Than, Equal To, etc)*
- **Text Length** *(Between, Less Than, Equal To, etc)*
- **Custom** *(Formula-Driven)*



FUN FACT: You can customize your own **error alerts**!



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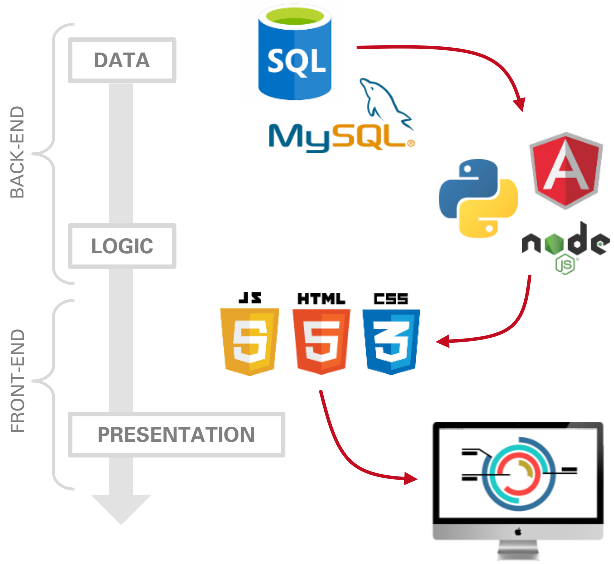
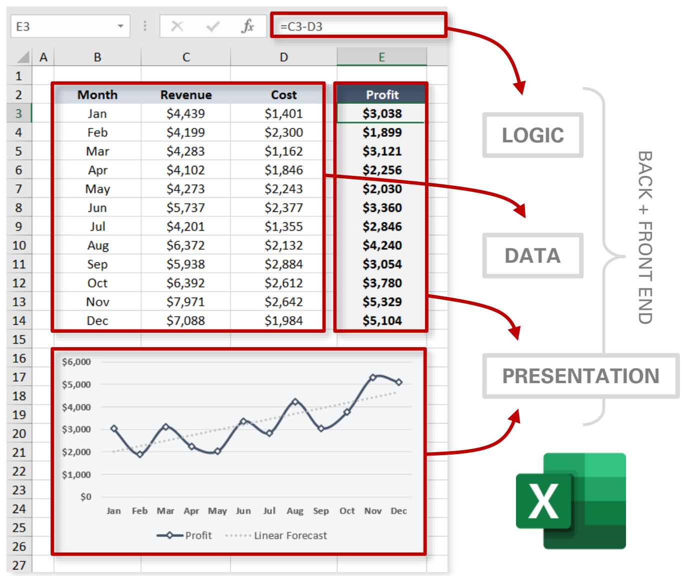
By definition, Excel is a **full-stack development** platform*; but rather than separating each layer of the process *(data, logic & presentation)*, Excel mixes them all within the same user interface:



VS



Month	Revenue	Cost	Profit
Jan	$4,439	$1,401	$3,038
Feb	$4,199	$2,300	$1,899
Mar	$4,283	$1,162	$3,121
Apr	$4,102	$1,846	$2,256
May	$4,273	$2,243	$2,030
Jun	$5,737	$2,377	$3,360
Jul	$4,201	$1,355	$2,846
Aug	$6,372	$2,132	$4,240
Sep	$5,938	$2,884	$3,054
Oct	$6,392	$2,612	$3,780
Nov	$7,971	$2,642	$5,329
Dec	$7,088	$1,984	$5,104

MAVEN ANALYTICS

*This is NOT a claim that Excel is always the **right** full-stack dev tool (or, in may cases, even a viable one). Rather, it's an effort to inspire users to think creatively about what Excel can do*